HealthConnect Week 8 Final Presentation Script

5-10 minute individual presentation | Data Science Track

Slide 1 - Introduction (about 45 seconds)

Good day. My name is Takudzwa Jeffrey Gumbo, and I am on the Data Science track in the AnalystLab Africa Experience Lab. My HealthConnect contribution focused on testing whether the available Rwanda healthcare data could support a useful predictive model while maintaining appropriate validation and feature governance. The overall HealthConnect project aims to improve appointment attendance and patient support, but an important limitation is that the supplied Rwanda appointment data does not contain an attendance or no-show outcome. I therefore did not fabricate that target. My Data Science work continued with the actual hospital-admission target available in the clinical transactions data.

Slide 2 - Project journey and Week 8 objective (about 50 seconds)

My work progressed from the broader Week 6 predictive analytics project into systematic Week 7 model testing and refinement, and finally into Week 8 model selection and documentation. The Week 8 goal is not to create another model for the sake of it. It is to decide what the evidence supports, document the final modelling outcome, connect it with the wider HealthConnect solution, and clearly state what the model can and cannot do.

Slide 3 - Data and target (about 50 seconds)

The final modelling dataset contains 5,000 integrated appointment-level records. The actual admission target comes from the Admissions sheet in the clinical transactions workbook, linked using AppointmentID. There are 586 admitted cases and 4,414 non-admitted cases, so the positive rate is about 11.7 percent. This imbalance means accuracy by itself is not a suitable success metric, so I focus on ROC-AUC, precision-recall performance, recall, balanced accuracy and cross-validation.

Slide 4 - Cross-track collaboration and feature decisions (about 75 seconds)

For Week 7 and final Week 8 integration, I collaborated with Agu Precious from the Data Analytics track. She reviewed my feature set against her validated analytics findings. Based on that review, I defined the prediction point as patient arrival before initial clinical assessment. I recalculated age as of the appointment date, added time-safe previous-visit count, month and weekday, retained facility and department IDs with caveats, and excluded Diagnosis and Severity because their timing could introduce data leakage. This collaboration did not just produce a discussion record; it directly changed the Data Science model design.

Slide 5 - Final model evaluation (about 75 seconds)

The final documented candidate is Gradient Boosting using the leakage-safer pre-assessment feature set. On the holdout partition, ROC-AUC is approximately 0.519 and precision-recall AUC is approximately 0.120. Five-fold cross-validated ROC-AUC is approximately 0.524 with a standard deviation of 0.025. At the default threshold, the confusion matrix contains 882 true negatives, one false positive, 117 false negatives and zero true positives. This is the most important result: despite a high-looking accuracy of 88.2 percent, the model fails to identify admitted patients at the default threshold.

Slide 6 - Baseline, refinement and final decision (about 70 seconds)

The original Week 6 Gradient Boosting candidate had ROC-AUC of about 0.497. During Week 7, one refinement produced a holdout ROC-AUC around 0.542, but cross-validation fell to around 0.487, so I rejected that apparent improvement as unstable. The final cross-track feature set improves methodology and gives ROC-AUC around 0.519 on holdout and 0.524 under cross-validation, but that is still close to chance. Therefore my final decision is not to deploy the model. A weak model should not be made operational simply because a project requires a final output.

Slide 7 - Business suitability and limitations (about 65 seconds)

The final model should not be used for clinical triage, patient prioritization, staffing allocation or automated care decisions. It is retained only as a reproducible analytical artifact showing the full modelling and validation process. Key limitations include weak discrimination, class imbalance, the lack of a fresh untouched final test set, limited verified pre-admission predictors, and the absence of an attendance or no-show target in the Rwanda appointment data.

Slide 8 - How Data Science fits the HealthConnect solution (about 60 seconds)

Within the wider HealthConnect solution, Data Analytics provides validated findings and data-quality context, Data Science evaluates whether those signals support prediction, ML Engineering can use the documented schema and preprocessing requirements for technical integration, Generative AI can support administrative interactions within its approved knowledge base, and Project Management coordinates readiness and limitations. My contribution is therefore not a production model; it is a validated modelling decision that prevents the wider solution from relying on an unreliable predictive component.

Slide 9 - Recommendations and closing (about 55 seconds)

My recommendations are to collect an explicit attendance or no-show outcome if missed appointments remain the main business objective, obtain stronger predictors that genuinely exist before the prediction point, freeze a new untouched test set, and define the operational costs of false positives and false negatives before threshold selection. My main learning is that responsible Data Science is not about making every model look successful. It is about testing assumptions, using cross-track evidence, recognizing limitations and making the decision supported by the data. Thank you.